FYS-STK3155/4155 Nov 2

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Building our intuition about RNNs ($m = 1$)

$$\frac{d^2x}{dt^2} + \eta \frac{dx}{dt} + x(t) = F(t)$$

acceleration — velocity dep. — external force

$$v(t) = \frac{dx}{dt}$$

$$\frac{dv}{dt} = -\eta \cdot v(t) - x(t) + F(t)$$

$$x(t_0) = x_0 \quad \wedge \quad v(t_0) = v_0$$

Discretize

$x \Rightarrow x_i = x(t_i)$

$v \Rightarrow v_i = v(t_i)$

$t_i = t_0 + \Delta t \cdot i \qquad i = 0, 1, 2, \ldots n$

$\Delta t = \dfrac{t_n - t_0}{n}$

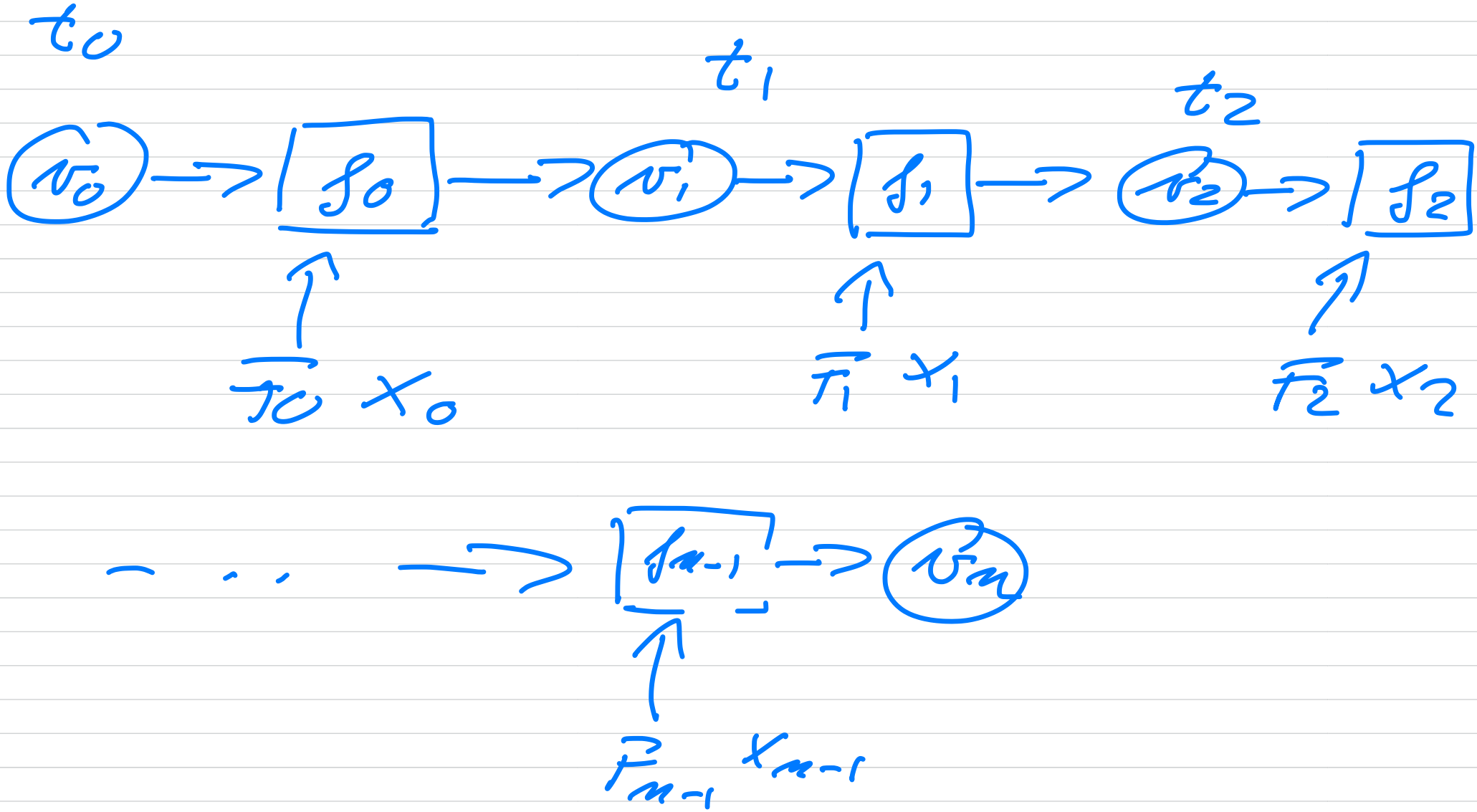
Euler's method

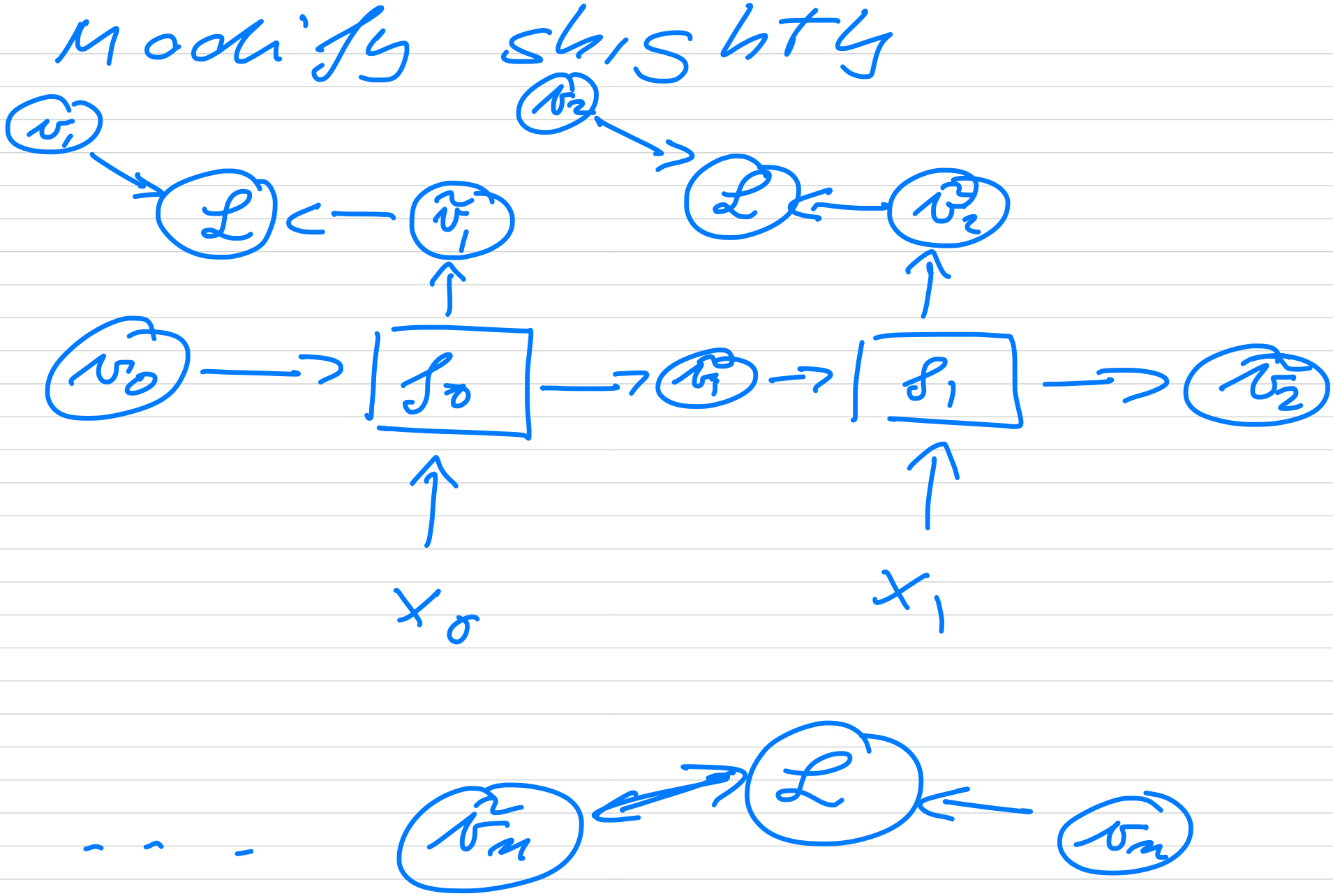
$x_{i+1} = x_i + \Delta t \cdot v_i$

$v_{i+1} = v_i + \underbrace{\Delta t \cdot (F_i - \eta v_i - x_i)}_{f(v_i, x_i, F_i, \Delta t)}$

$= v_i + f_i$

Graphical representation




modify slightly
v_1
$\mathcal{L}$
$\tilde{v}_1$
v_2
$\mathcal{L}$
v_2
v_0
f_0
v_1
f_1
v_2
x_0
x_1
$\tilde{v}_m$
$\mathcal{L}$
v_m

More general

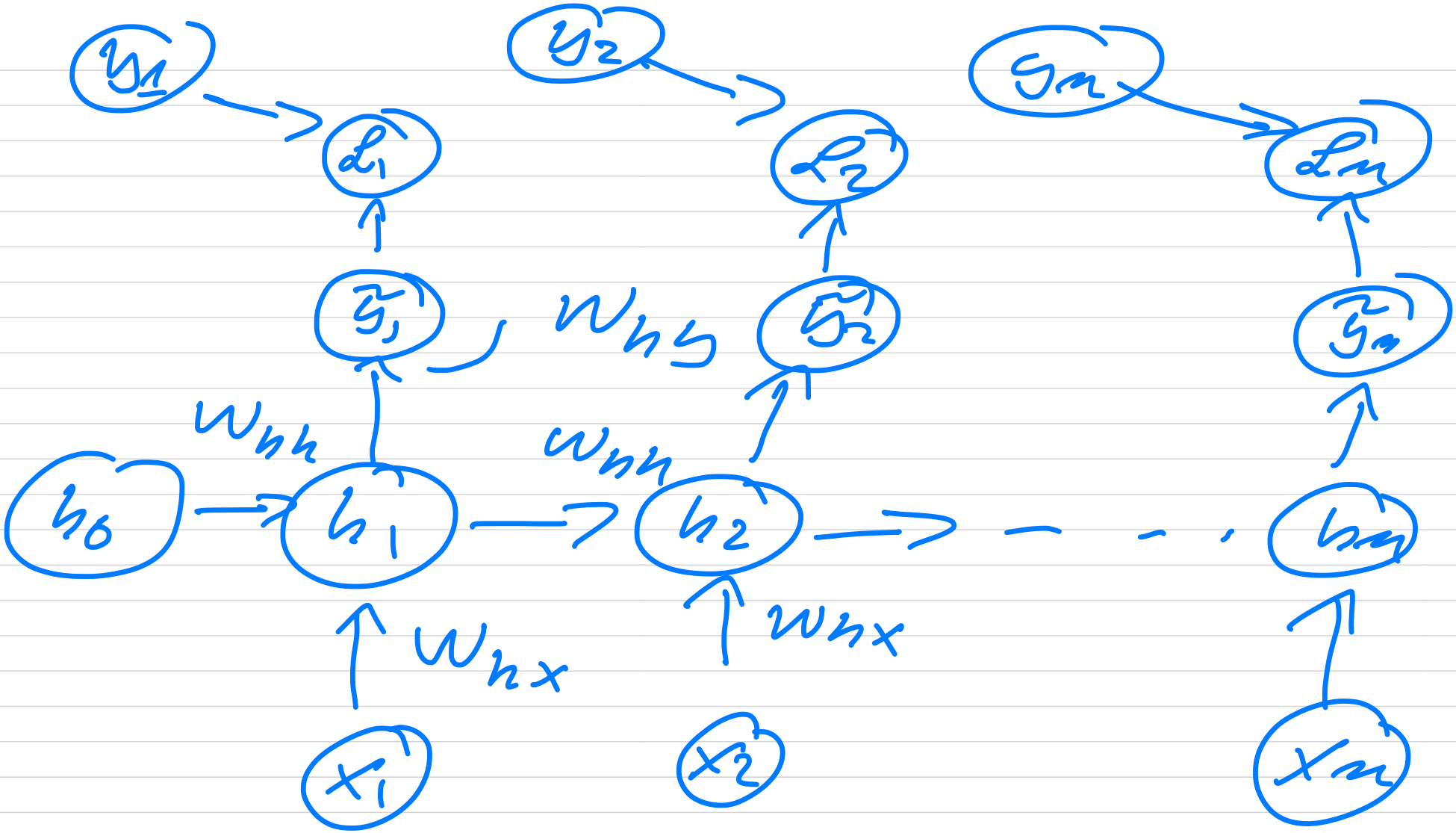
$v_i \rightarrow y_i$ (Target / output)

$\tilde{v}_i \rightarrow \tilde{y}_i$ model output at t_i

$f_i \rightarrow \underbrace{h_i(x_i, h_{i-1})}_{\text{hidden cages}}$

introduce weights

W_{hh}, W_{hy}, W_{hx}


y_1
y_2
y_m
L_1
L_2
L_m
$\hat{y}_1$
$\hat{y}_2$
$\hat{y}_m$
W_{hy}
W_{hh}
W_{hh}
h_0
h_1
h_2
h_m
W_{hx}
W_{hx}
x_1
x_2
x_m

Many - to many

parameters

$$\theta = \{ W_{hh}, W_{hx}, W_{hy}, b, c \}$$

$$\tilde{y}_i = \phi(W_{hy} h_i + c_i)$$

$$h_i = \sigma(W_{hh} h_{i-1} + W_{hx} x_i + b_i)$$

h_i = value of hidden layer

ϕ and σ are activation functions

$$\mathcal{L}(\theta) = \sum_{n=0}^{m} \mathcal{L}_n(\tilde{y}_n, y_n; \theta)$$

Back propagation in time

$\nabla_c \mathcal{L}$, $\nabla_b \mathcal{L}$, $\nabla_{W_{hh}} \mathcal{L}$,

$\nabla_{W_{hx}} \mathcal{L}$, $\nabla_{W_{hy}} \mathcal{L}$

Training a basic (simple) RNN can be difficult due to long-range information can lead to exploding gradients

Exploding gradients and gradient clipping

$W_{hh} \rightarrow W$

$$\frac{\partial L_t}{\partial W} = \frac{\partial L_t}{\partial \tilde{y}_t} \frac{\partial \tilde{y}_t}{\partial h_t} \frac{\partial h_t}{\partial W}$$

$$L_t = \frac{1}{2} \| (\tilde{y}_t - y_t) \|_2^2$$

$$\frac{\partial h_t}{\partial W} = \sum_{k=0}^{n-1} \frac{\partial h_t}{\partial h_k} \frac{\partial h_k}{\partial W}$$

$$\frac{\partial h_i}{\partial h_k} = \prod_{i=k+1}^{n} \frac{\partial h_i}{\partial h_{i-1}}$$

computed at a multiplication of adjacent "time" steps

w (weight sharing) is the same for all "time" steps

$$h_i = w \cdot h_{i-1}$$

$$h_i = w \cdot w \cdot \ldots \cdot w \cdot h_0 = w^i h_0$$

assume W can be diagonalized, with eigenpairs (λ_i, w_i)

$$h_0 = \sum_i \alpha_i w_i$$

$$W \cdot h_0 = \sum_i \alpha_i W w_i$$

$$= \sum_i \alpha_i \lambda_i w_i$$

$$W^t h_0 = \sum_i \alpha_i \lambda_i^t w_i$$

$$\lambda_0 > \lambda_1 > \lambda_2 > \cdots > \lambda_m$$

$$\lim_{t \to \infty} w^t h_0 \approx \lambda_0^t \alpha_0 v_0$$

$\lambda_0 > 1$, then the derivative may grow quickly $\Rightarrow$ exploding gradients which leads to gradient clipping

$\vec{g}$ = gradient

if $\|\vec{g}\|_2 > \varepsilon$ (fixed)

$$\vec{g} \leftarrow \frac{\varepsilon}{\|\vec{g}\|_2} \vec{g}$$